Analytical Chemistry Cumulative Exam
February 17, 2012

1. Describe how a deuterium lamp can be used to provide background correction for an atomic absorption spectrum.

2. A 5.00-ml sample of blood was treated with trichloroacetic acid to precipitate proteins. After ultracentrifugation, the resulting solution was brought to pH 3 and extracted with two 5-ml portions of methyl isobutyl ketone containing the organic lead complexing agent APCD. The extract was aspirated directly into an air-acetylene flame yielding an absorbance of 0.444 at 283.3 nm. Five-milliliter aliquots of standard solutions containing 0.250 and 0.450 ppm Pb were treated in the same way and yielded absorbances of 0.396 and 0.599. Calculate the concentration of Pb (ppm) in the blood sample, assuming that Beer's law is followed.

3. An analyst attempts to determine strontium with an atomic absorption instrument equipped with a nitrous-oxide-acetylene burner, but the sensitivity associated with the 460.7 nm atomic resonance line is not satisfactory. Suggest at least three things that might be tried to increase sensitivity.

4. Why is an electrothermal analyzer more sensitive than a flame atomizer?

5. Some spectrofluorimeters are capable of providing both excitation and emission spectra. How is this accomplished? A simple clearly labeled instrument diagram could be used to answer this question.